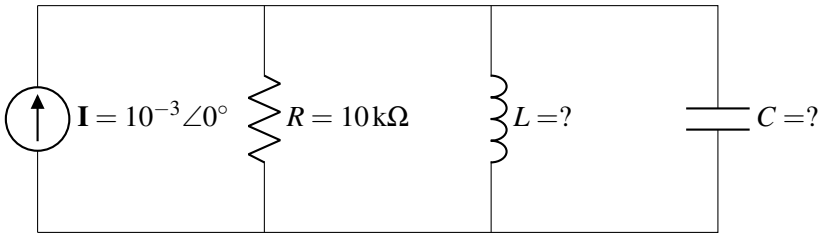
CSM
Fall 2025

EECS 16B
Worksheet 7

1. Parallel Resonant Circuit

(a) Find the L and C values for a parallel resonant circuit that has $R = 10\,\text{k}\Omega$, $f_0 = 1\,\text{MHz}$, and $B = 100\,\text{kHz}$. You are also told that $\mathbf{I} = 10^{-3}\angle 0^\circ$ A.

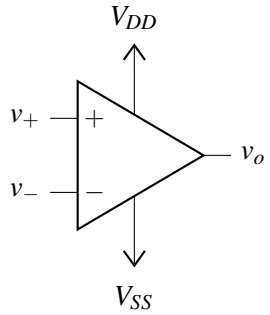


(b) Calculate the current phasors through each of the elements at resonance.

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2. OpAmps as Comparators

(a) Now you've learned about a new component, the operational amplifier. What can this circuit component be used for? How does it work?

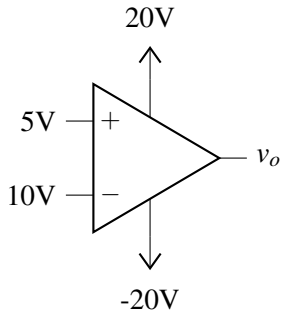


(b) Above we have the basic schematic of the op-amp. What does each terminal mean? How do we determine the output voltage? Are there any constraints?

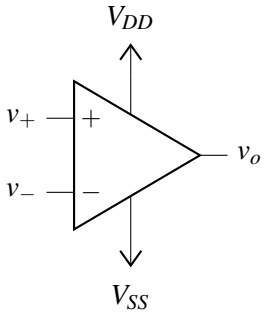
(c) Much of EECS 16B will have our analysis be restricted to ideal op-amps. However, it is important to know non-ideal behaviors. What are some main differences between ideal and non-ideal op-amps?

(d) Below are some op-amp configurations. Please identify the output voltage V_o for each.

i.



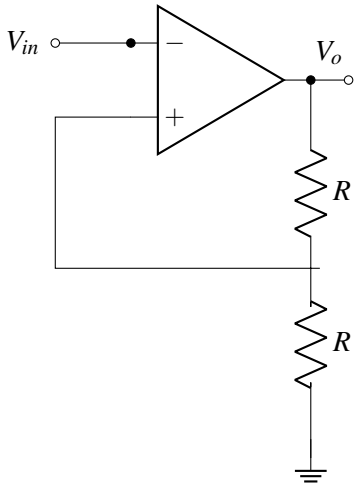
ii.



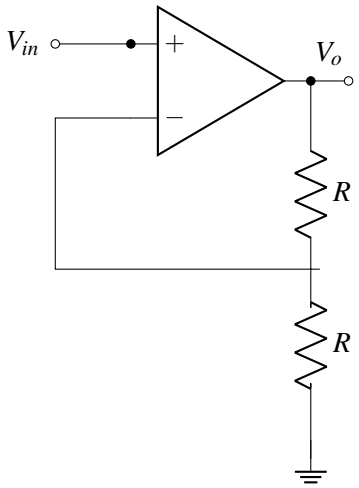
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3. OpAmps as Amplifiers

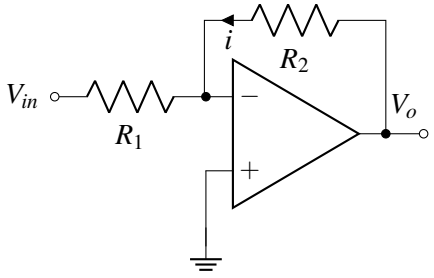
(a) In general, what is feedback? What are positive and negative feedback? And how do you introduce feedback into op-amps?



(b) Referring to the above example of an op-amp employing positive feedback, start with the assumption that $V_+ = V_-$. Now, taking into account the fact that $V_{out} = A(V_+ - V_-)$ (where A is some gain), try to raise V_{out} (by messing with V_{in}) and observe what happens. Try to lower V_{out} and observe that too.

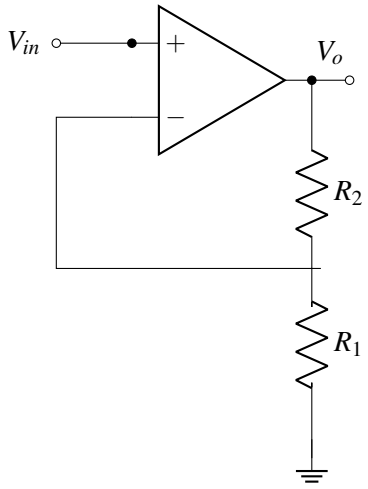


(c) Now do the same with this above example of an op-amp that uses negative feedback



(d) In terms of the input V_{in}, what is V_{out} for this circuit? What is the gain of this circuit?

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(e) Again, in terms of the input V_{in}, what is V_{out} for this circuit? What is the gain?

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